

Support Vector Machine (SVM)

Introduction:Support Vector Machine (SVM) is a Supervised Machine Learning Algorithm.

It is used for:

- Classification Problems
- Regression Problems

Since SVM is a supervised learning algorithm, it requires labeled data for training.

Example

- Email Spam vs Not Spam
- Cat vs Dog
- Positive vs Negative Reviews

The main objective of SVM is to separate different classes of data as accurately as possible.

Hyperplane:The most important concept in SVM is the Hyperplane.

Definition:A hyperplane is a decision boundary that separates data points belonging to different classes.

For example:

+ + + +
+ + + +

----- ← Hyperplane

- - - -
- - - -

The line dividing the positive (+) and negative (-) classes is called the Hyperplane.

Purpose of Hyperplane

- Separates data into different classes.
- Helps classify new data points.
- Acts as a decision boundary.

If a new point falls on one side of the hyperplane, it belongs to one class; if it falls on the other side, it belongs to another class.

Mathematical Representation of Hyperplane: The equation of a hyperplane is:

$$w^T x + b = 0$$

Where:

- w = Weight Vector
- x = Feature Vector
- b = Bias (Intercept)

Multiple Features: If there are multiple features:

$$w_1 x_1 + w_2 x_2 + \dots + b = 0$$

Where:

- (x_1, x_2) = Input Features
- (w_1, w_2) = Corresponding Weights

Example of a Hyperplane:

Suppose we want to classify animals as:

- Cat = +1
- Dog = -1

Using:

- Weight
- Height

After training, SVM may generate:

$$[3 \times \text{Weight} + 2 \times \text{Height} - 50 = 0]$$

Here:

- Weight coefficient = 3
- Height coefficient = 2
- Bias = -50

This equation becomes the decision boundary that separates cats from dogs.

Margin: Margin is the distance between the hyperplane and the nearest data points from each class.

Positive Class

+ +
+ +

----- Margin Line

===== Hyperplane

----- Margin Line

- -
- -

Negative Class

The distance between:

- Hyperplane
- Closest data points

is called the Margin.

Why Margin is Important: SVM always tries to maximize the margin.

Reason

A larger margin:

- Provides better separation between classes.
- Improves model generalization.
- Reduces classification errors.
- Makes the model more robust.

Comparison

Small Margin

+ + +

=====

- - -

Poor separation.

Large Margin

+ + +

=====

- - -

Better separation, therefore SVM always aims to maximize the margin.

Support Vectors: Support Vectors are the data points closest to the hyperplane.

+ +

+ ● ← Support Vector

===== Hyperplane

● - ← Support Vector

- -

These points determine:

- Position of the hyperplane
- Orientation of the hyperplane
- Margin size

Importance

Without support vectors, SVM cannot determine the optimal separating boundary.

In fact, the algorithm gets its name from these important points:

Support Vector Machine

Hard Margin SVM: Hard Margin SVM assumes that the data is perfectly linearly separable.

+ + + +

===== Hyperplane

- - - -

Characteristics:

- No overlap between classes.
- Perfect separation.
- All points lie outside the margin boundaries.

Advantage:

Simple and accurate when data is perfectly separable.

Limitation:

Real-world data is rarely perfectly separable.

Soft Margin SVM: Soft Margin SVM allows some overlap between classes.

Example:

+ + +

+ - +

===== Hyperplane

- - -

Some points may lie on the wrong side of the boundary.

Why Use Soft Margin?

Real-world datasets often contain:

- Noise
- Outliers
- Overlapping classes

Soft Margin SVM handles these situations effectively.

Linear Separability:

A dataset is linearly separable if a straight line can divide the classes.

Example

+ + + +

- - - -

Classes are clearly separated.

This is called Linear Separation.

Non-Linear Data:

Sometimes data cannot be separated using a straight line.

Example:

+ + +

+ -

- + -

+ +

No single straight line can separate the classes.

This is called Non-Linear Data.

Kernel Functions: To handle non-linear data, SVM uses Kernel Functions.

A kernel function transforms data into a higher-dimensional space where separation becomes easier.

Idea

Convert:

- 2D Data → 3D Data
- 3D Data → Higher Dimensions

After transformation, the data may become linearly separable.

Types of Kernel Functions:

A. Linear Kernel: Used when data is already linearly separable.

Advantages

- Fast
- Simple
- Efficient

B. Polynomial Kernel: Used when data has curved relationships.

Advantages

- Captures complex patterns
- Works for moderately non-linear datasets

C. Radial Basis Function (RBF) Kernel:

One of the most popular kernels.

Advantages

- Handles highly non-linear data
- Very flexible
- Widely used in practical applications

Key Points to Remember:

1. SVM is a supervised learning algorithm.
2. It is used for classification and regression.
3. Hyperplane acts as a decision boundary.
4. Margin is the distance between the hyperplane and nearest data points.
5. SVM always maximizes the margin.
6. Support vectors are the closest points to the hyperplane.
7. Hard Margin assumes perfect separation.
8. Soft Margin allows overlap between classes.
9. Kernel functions handle non-linear data.
10. Common kernels:
 - Linear Kernel
 - Polynomial Kernel
 - RBF Kernel